

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

Course Name: Cloud Computing and Big Data Analytics

Course Code: CSA15

Course Outcome Covered:

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Question 1

A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering.

Tasks:

- Design a cloud architecture using scalability and caching techniques.
- Explain how load balancing and auto-scaling can ensure smooth streaming.
- Discuss cost optimization strategies for such high-demand workloads.

Answer

Introduction

Cloud computing enables media companies to stream videos efficiently by using scalable infrastructure, content caching, and automated resource management.

Proposed Cloud Architecture

- Content Delivery Network (CDN): Delivers video content from edge servers closest to users, reducing latency.
- Load Balancer: Distributes user requests evenly across multiple application servers.

- Auto-Scaling Group: Automatically adds or removes servers based on traffic demand.
- Object Storage: Stores video files securely and durably.
- Database Service: Stores user accounts, subscriptions, and viewing history.
- Monitoring Service: Tracks CPU, bandwidth, and application health.

Load Balancing and Auto-Scaling

- Load Balancer
 - Prevents server overload.
 - Improves response time.
 - Ensures continuous service availability.
- Auto-Scaling
 - Launches additional instances during peak traffic.
 - Removes unused instances during low demand.
 - Maintains consistent streaming performance.

Cost Optimization Strategies

- Use Reserved Instances for predictable workloads.
- Use Spot Instances for background processing.
- Store frequently accessed videos in CDN caches.
- Archive old videos using low-cost cloud storage.
- Enable automatic scaling to avoid paying for idle resources.

Conclusion

By combining CDN, load balancing, auto-scaling, and cost-efficient storage, the media company can deliver smooth, high-quality streaming while minimizing operational costs.

Question 2

A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments.

Tasks:

- Propose a solution using container orchestration and multi-cloud strategies.
- Explain how service discovery and scaling can be handled efficiently.
- Highlight the benefits and risks of a multi-cloud deployment model.

Answer

Introduction

Containers provide lightweight application deployment, while orchestration platforms manage containerized applications across different cloud environments.

Proposed Solution

- Deploy applications using Docker containers.
- Manage containers with Kubernetes across multiple cloud providers.
- Use a centralized Container Registry for consistent application images.
- Implement Infrastructure as Code (IaC) for automated deployments.

Service Discovery

- Kubernetes automatically assigns service names and internal DNS.
- Applications communicate without knowing server IP addresses.
- Load balancing routes traffic to healthy container instances.

Scaling

- Horizontal Pod Autoscaler (HPA) increases or decreases container replicas based on CPU or memory usage.
- Cluster Autoscaler adds or removes worker nodes when required.

Benefits of Multi-Cloud

- High availability.
- Reduced vendor lock-in.
- Better disaster recovery.
- Improved global performance.
- Flexibility in selecting cloud services.

Risks

- Increased management complexity.
- Higher operational costs.
- Security policy inconsistencies.
- Data synchronization challenges.

Conclusion

Kubernetes with container orchestration provides consistent deployment, efficient scaling, and simplified management while enabling resilient multi-cloud operations.

Question 3

An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Tasks:

- Design a solution using cloud-native backup, replication, and versioning.
- Explain how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are achieved.
- Provide steps to ensure business continuity during failures.

Answer

Introduction

Backup and disaster recovery protect cloud data against accidental deletion, cyberattacks, and system failures.

Proposed Backup Solution

- Enable Cloud Backup for databases and storage.
- Use Object Versioning to restore deleted or modified files.
- Configure Cross-Region Replication for disaster recovery.
- Schedule automatic daily backups.
- Store backup copies in geographically separate regions.

Achieving RTO and RPO

Recovery Time Objective (RTO)

- Fast restoration using automated recovery tools.
- Standby systems reduce downtime.

Recovery Point Objective (RPO)

- Frequent incremental backups.
- Real-time data replication minimizes data loss.

Business Continuity Steps

1. Enable automated backups.
2. Replicate critical data to another region.
3. Regularly test disaster recovery plans.
4. Monitor backup health and integrity.
5. Train employees on recovery procedures.

Conclusion

Cloud-native backups, replication, and versioning ensure fast recovery, minimal data loss, and uninterrupted learning services during failures.

Question 4

A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements.

Tasks:

- Design a hybrid architecture that ensures secure communication between environments.
- Explain how identity management and encryption should be implemented.
- Discuss challenges in managing hybrid cloud environments.

Answer

Introduction

Hybrid cloud combines private infrastructure with public cloud resources, allowing organizations to meet security, compliance, and scalability requirements.

Proposed Hybrid Architecture

- Sensitive applications remain in the on-premises datacenter.
- Non-sensitive applications are deployed in the public cloud.
- Establish a secure VPN or dedicated private connection between environments.
- Use centralized monitoring and management tools.

Identity Management

- Implement Single Sign-On (SSO).
- Use Role-Based Access Control (RBAC).
- Enable Multi-Factor Authentication (MFA).
- Synchronize identities across cloud and on-premises systems.

Encryption

- Encrypt data in transit using TLS/SSL.
- Encrypt stored data using AES-256.
- Use centralized key management services.

Challenges

- Complex infrastructure management.
- Compliance and regulatory requirements.
- Data synchronization issues.
- Increased operational complexity.
- Higher maintenance costs.

Conclusion

A secure hybrid cloud architecture with strong identity management, encryption, and centralized administration provides compliance, flexibility, and enhanced security.

Question 5

A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down.

Tasks:

- Propose a multi-region deployment strategy in the cloud.
- Explain how data replication and failover mechanisms should be configured.
- Discuss monitoring and alerting strategies for proactive issue detection.

Answer

Introduction

High availability and fault tolerance ensure cloud applications remain operational despite hardware failures, network outages, or regional disasters.

Multi-Region Deployment Strategy

- Deploy application servers in two or more cloud regions.
- Use a Global Load Balancer to distribute user requests.
- Maintain identical application environments in all regions.
- Use DNS-based traffic routing for regional access.

Data Replication

- Enable synchronous replication within the same region for consistency.
- Configure asynchronous replication across regions for disaster recovery.
- Schedule regular backup verification.

Failover Configuration

- Automatically redirect traffic to healthy regions.
- Continuously monitor application health.
- Use automated recovery scripts for rapid restoration.

Monitoring and Alerting

- Monitor CPU, memory, storage, network, and application health.
- Configure alerts for high latency, service failures, and resource exhaustion.
- Use centralized logging and dashboards for real-time visibility.
- Perform regular health checks and failover testing.

Benefits

- High availability.
- Improved fault tolerance.
- Reduced downtime.
- Better user experience.
- Faster disaster recovery.

Conclusion

A multi-region cloud architecture with global load balancing, data replication, automated failover, and proactive monitoring ensures continuous application availability and reliable service even during regional failures.